Project Title

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https://github.com/hourlilies/elec5305-project-550790019

I. Project Overview

This project aims to develop a synthesizer, sampler, and music sequencer for any device capable of running a modern web browser supporting WebAudio, using the emscripten compiler toolchain that turns C++ DSP code into WebAssembly. The goal is to make music production more accessible, because pretty much any computer, whether that be a smartphone, laptop, or even a single-board computer, can run a modern web browser that supports WebAudio. This is important because anyone should be able to play with and make music without needing to install and learn a comprehensive digital audio workstation, designed for professional musicians and not the general public.

Though the advent of generative AI in music creation has lowered the barrier of entry tremendously, empowering authors to make and tune music to their taste with just a few text prompts and a bit of luck, it has come at the cost of abstracting away the underlying signal processing and audio intuitions that makes it all work. This project aims to expose all of that, the wires, knobs, and switches of music making, with the hopes of allowing the prospective musician to see and poke at the machine that makes modern music tick, whose human invention and decades of production is what allows powerful GenAI tools like Suno or Udio to even exist.

II. Background and Motivation

Despite its relatively recent history, with the first publication of a working draft in only 2011[1], the WebAudio API has seen extensive development over the course of a decade, eventually receiving an endorsement from the W3C in 2021 with a Recommendation[2], and implemented in most major browsers with full support since at least 2015[3] (with the exception of Safari, which added full support only in 2021).

Of course, audio has always played an important role on the internet. As Boris Smus describes in their summary on the brief history of audio on the web[**?**]

III. Proposed Methodology

methodology.tex

IV. Expected Outcomes

outcomes.tex

V. Timeline

timeline.tex

VI. References

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VII. Appendix

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[1] https://www.w3.org/TR/2011/WD-webaudio-20111215/
[2] https://www.w3.org/TR/webaudio-1.0/
[3] https://developer.mozilla.org/en-US/docs/Web/API/Web_Audio_API#browser_compatibility